3905. Multi Source Flood Fill

Solved

Medium Topics Hint

You are given two integers `n` and `m` representing the number of rows and columns of a grid, respectively.

You are also given a 2D integer array `sources`, where `sources[i] = [$r_i$, $c_i$, $color_i$]` indicates that the cell `($r_i$, $c_i$)` is initially colored with `$color_i$`. All other cells are initially uncolored and represented as 0.

At each time step, every currently colored cell spreads its color to all adjacent **uncolored** cells in the four directions: up, down, left, and right. All spreads happen simultaneously.

If **multiple** colors reach the same uncolored cell at the same time step, the cell takes the color with the **maximum** value.

The process continues until no more cells can be colored.

Return a 2D integer array representing the final state of the grid, where each cell contains its final color.

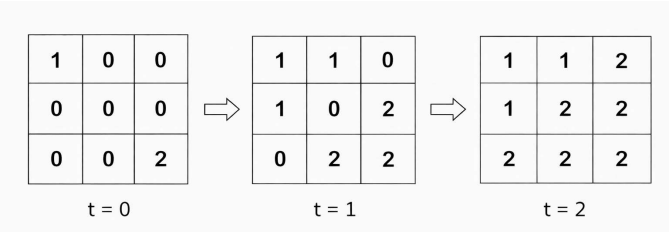
Example 1:

Input: n = 3, m = 3, sources = [[0,0,1],[2,2,2]]

Output: [[1,1,2],[1,2,2],[2,2,2]]

Explanation:

The grid at each time step is as follows:



At time step 2, cells `(0, 2)`, `(1, 1)`, and `(2, 0)` are reached by both colors, so they are assigned color 2 as it has the maximum value among them.

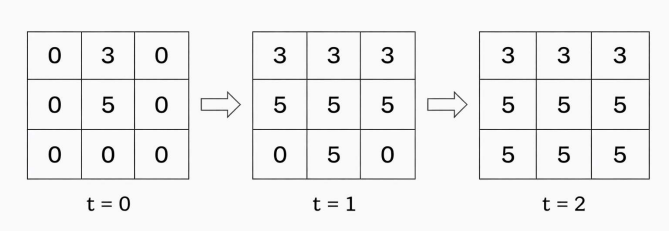
Example 2:

Input: n = 3, m = 3, sources = [[0,1,3],[1,1,5]]

Output: [[3,3,3],[5,5,5],[5,5,5]]

Explanation:

The grid at each time step is as follows:



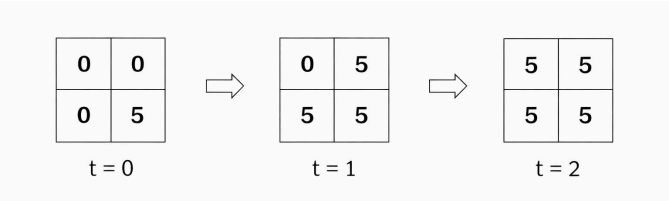
Example 3:

Input: n = 2, m = 2, sources = [[1,1,5]]

Output: [[5,5],[5,5]]

Explanation:

The grid at each time step is as follows:



Since there is only one source, all cells are assigned the same color.

Constraints:

- $1 <= n, m <= 10^5$
- $1 <= n * m <= 10^5$
- $1 <= sources.length <= n * m$
- $sources[i] = [r_i, c_i, color_i]$
- $0 <= r_i <= n - 1$
- $0 <= c_i <= m - 1$
- $1 <= color_i <= 10^6$
- All (r_i, c_i) in $sources$ are distinct.

Seen this question in a real interview before? 1/6

Yes No

Accepted 21,973 /39.4K | Acceptance Rate 55.8 %

Topics

Hint 1

Hint 2

Hint 3

Hint 4

Discussion (37)

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